

Generalization of $3n + 1$ Collatz Orbits via Jacobsthal Sequences: Visualization using Voronoi Tessellation in 2-adic Logarithmic Spiral Space

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Abstract

In this study, the behavior of natural numbers in the $3n + 1$ Collatz map is visualized by mapping them onto a 2-adic logarithmic spiral space and applying Voronoi tessellation, revealing geometric patterns hidden within the orbital structure. Particularly, within the local preimage tree centered at an arbitrary odd node M , it was confirmed that the following Jacobsthal (J)-type sequences appear repeatedly, suggesting their contribution to the topological partition of the natural number space.

- **Jacobsthal sequence (J):** J even terms (J_{even}) and J odd terms (J_{odd})
- **Triple J sequence ($3J$):** $3J_{even}$ and $3J_{odd}$
- **Conjugate J sequence (cJ):** cJ_{even} and cJ_{odd}

By visualizing these six sequences along with the $a \cdot 2^b$ sequence—totaling seven systems—as an RGB color topology, we demonstrate that natural numbers possess a highly ordered geometric arrangement. Note that this study is based on observational visualization and aims to present geometric patterns rather than rigorous mathematical proofs.

1. Introduction

The Collatz conjecture is known for the complexity of its orbital structures despite its simple operational rules. While conventional research has primarily focused on analysis over the 1D number line, this study maps and analyzes natural numbers in the following 2-adic logarithmic spiral space:

$$r = \log_2(n), \theta = 2\pi(r \bmod 1)$$

This mapping possesses the following characteristics, making it highly suitable for capturing the geometric properties of Collatz orbits:

- Multiplication by 2: Radial translation
- Odd operations: Angular phase jumps
- Preservation of 2-adic self-similarity

Furthermore, applying Voronoi tessellation on this continuous space allows for the stained-glass-like visualization of the basins occupied by natural numbers.

2. Generalized Jacobsthal (J) Sequences for Arbitrary Odd Roots (M)

The odd solutions of the inverse Collatz map generally take the following form:

$$n = \frac{2^k \cdot m - 1}{3}$$

To formalize this structure, we define the following scaling factor A for an arbitrary odd node M :

$$A = \begin{cases} M & (M \equiv 1 \pmod{3}) \\ 2M & (M \equiv 2 \pmod{3}) \\ M & (M \equiv 0 \pmod{3}) \end{cases} \quad (\text{Auxiliary definition})$$

Using this A , J -type sequences can be observed within sections of the inverse Collatz tree.

1) Jacobsthal Sequence (J)

$$J_n(M) = \frac{A \cdot 2^n - (-1)^n}{3}$$

2) Triple Jacobsthal Sequence ($3J$)

$$3J_n(M) = A \cdot 2^n - (-1)^n$$

3) Conjugate Jacobsthal (Lucas-type) Sequence (cJ)

$$cJ_n(M) = A \cdot 2^n + (-1)^n$$

By categorizing the index n of the above J -type sequences into even and odd, we classify them into the following six systems:

$$J_{\text{even}}, J_{\text{odd}}, 3J_{\text{even}}, 3J_{\text{odd}}, cJ_{\text{even}}, cJ_{\text{odd}}$$

Adding the direct powers-of-two orbit shown below, the local natural number space is entirely classified into seven systems.

$$M \cdot 2^b$$

3. Seven-System RGB Topology Visualization

These seven systems were visualized using color mappings based on the RGB primary colors.

System	Color
J_{even}	Light Blue
J_{odd}	Blue
$3J_{\text{even}}$	Pink
$3J_{\text{odd}}$	Orange-Red
cJ_{even}	Green
cJ_{odd}	Light Green
$M \cdot 2^b$	Yellow

Additionally, when multiple sequences intersect at the same natural number, a blended color representation is applied. The following composite colors were assigned:

- Purple ($J \times 3J$)
- Aquamarine ($J \times cJ$)
- Yellow-Green ($cJ \times 2^b$)

This color mapping allows for an intuitive understanding of how the natural number space is partitioned and where sequences intersect.

4. Verification at Initial Value 7

Generalized J sequences can be defined even when the initial terms are non-integers. In the context of the Collatz odd cores, only integer terms are discussed; fractional terms serve as auxiliary structures illustrating the sequence's linearity. A generalized J sequence with an initial term j_0 and a subsequent term j_1 is denoted as $J(j_0, j_1)$. The general form remains the following equation:

$$J_n = (A \cdot 2^n - (-1)^n)/3$$

The decomposition rule into the J -basis is as follows. When deriving a local generalized J sequence targeting an arbitrary odd number N , we treat N as J_2 based on the J recurrence relation $J_n = J_{n-1} + 2J_{n-2}$, decomposing it as follows:

$$N = j_2 = j_1 + 2 \cdot j_0$$

This uniquely determines the sequence $J(j_0, j_1)$ containing the target number.

As an example, we verified this by back-tracing the odd Collatz orbit of the initial value 7 (Figure 1):

$$7, 11, 17, 13, 5, 1$$

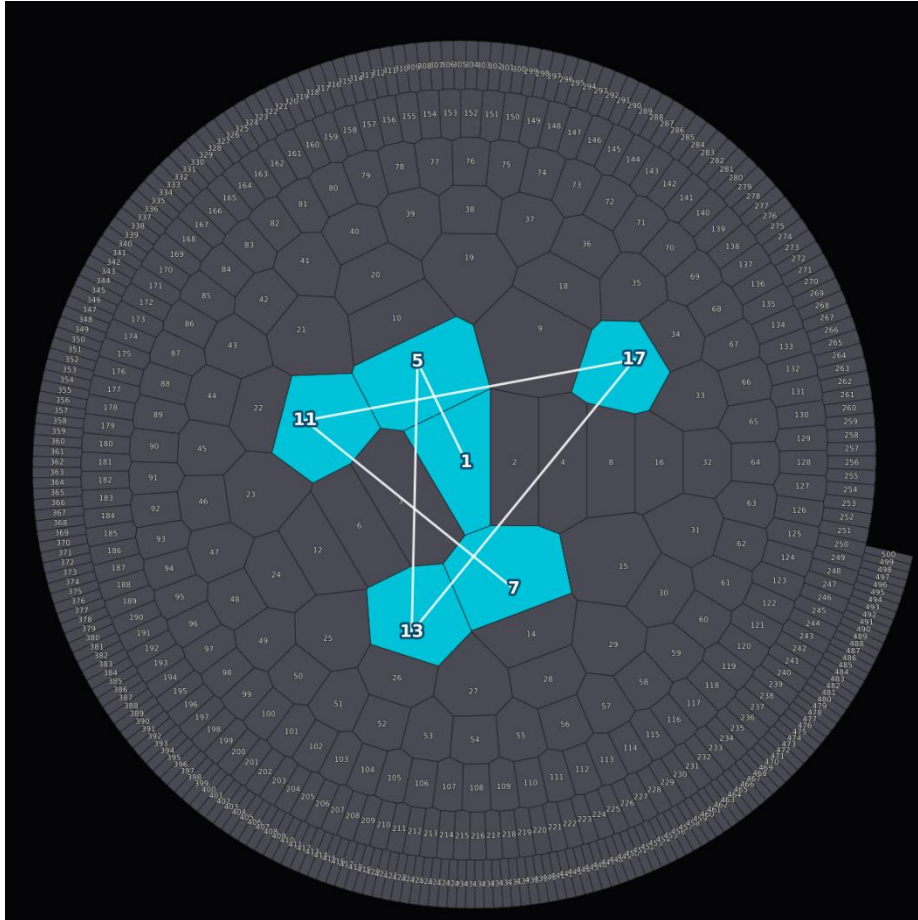


Figure 1. Transition of the odd Collatz orbit for the initial value 7 ($7 \rightarrow 11 \rightarrow 17 \rightarrow 13 \rightarrow 5 \rightarrow 1$). The trajectory is visualized by highlighting the regions (cells) occupied by the natural numbers on the orbit and connecting them with straight lines over the Voronoi tessellation in the 2-adic logarithmic spiral space.

4.1 Step 1: $5 \rightarrow 1$ (Landing on the Final Attractor $1 \cdot 2^b$)

First, we decompose 5 into the J -basis and generate the J sequence containing 5.

$$5 = 3 + 2 = 3 + 1 \cdot 2$$

$$J(1, 3) = 1, 3, 5, 11, 21, \dots$$

Next, we generate $3J$ and cJ .

$$3J = 3, 9, 15, 33, 63, \dots$$

$$cJ = 5, 7, 17, 31, 65, \dots$$

It can be observed that the $1 \cdot 2^b$ axis passes between $3J$ and cJ .

$$1 \cdot 2^b = 4, 8, 16, 32, 64, \dots = 2^2, 2^3, 2^4, 2^5, 2^6, \dots$$

The J_{even} terms of the 5-system (1, 5, 21, ...) land on the $1 \cdot 2^b$ axis via the $3n + 1$ operation. Since $(-1)^n = +1$ for J_{even} , the value after the $3 \times$ multiplication ($3J_{\text{even}}$) takes the form of $a \cdot 2^b - 1$, and the subsequent $+1$ operation shifts it onto the $a \cdot 2^b$ axis (**Figure 2**).

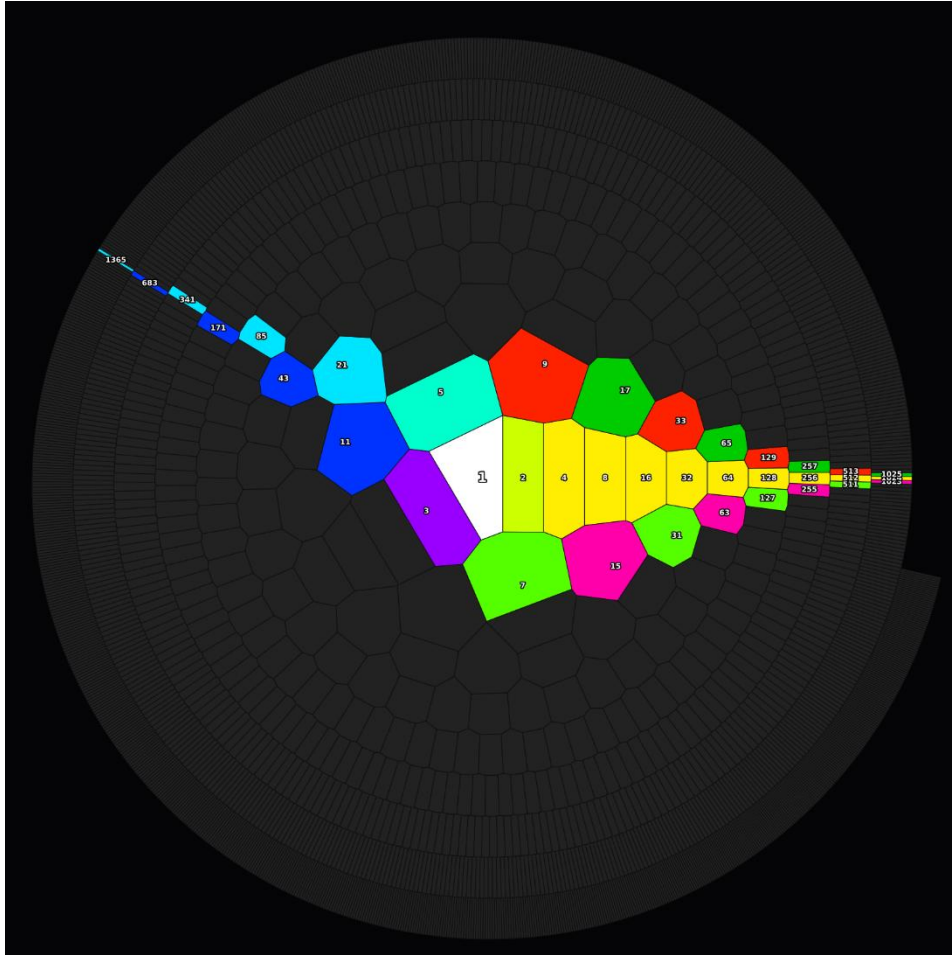


Figure 2. The $5 \rightarrow 1$ transition in Step 1 and the seven-system topology around the odd root $M = 1$.

It illustrates the geometric structure where the J_{even} sequence containing 5 (1, 5, 21, ...) lands on the $1 \cdot 2^b$ axis—located between $3J$ and cJ —via the $3n+1$ operation, visualized by the RGB stained-glass color mapping.

4.2 Step 2: $13 \rightarrow 5$ (Transition to the $5 \cdot 2^b$ Axis)

Similarly, we generate the J sequence group containing 13.

$$13 = 7 + 6 = 7 + 3 \cdot 2$$

$$J(3, 7) = 3, 7, 13, 27, 53, \dots$$

$$3J = 9, 21, 39, 81, 159, \dots$$

$$cJ = 11, 19, 41, 79, 161, \dots$$

$$5 \cdot 2^b = 10, 20, 40, 80, 160, \dots = 5 \cdot 2^1, 5 \cdot 2^2, 5 \cdot 2^3, 5 \cdot 2^4, 5 \cdot 2^5, \dots$$

Therefore, the J_{even} terms of the 13-system $(3, 13, 53, \dots)$ transition to the $5 \cdot 2^b$ axis *via* the $3n + 1$ operation (**Figure 3**).

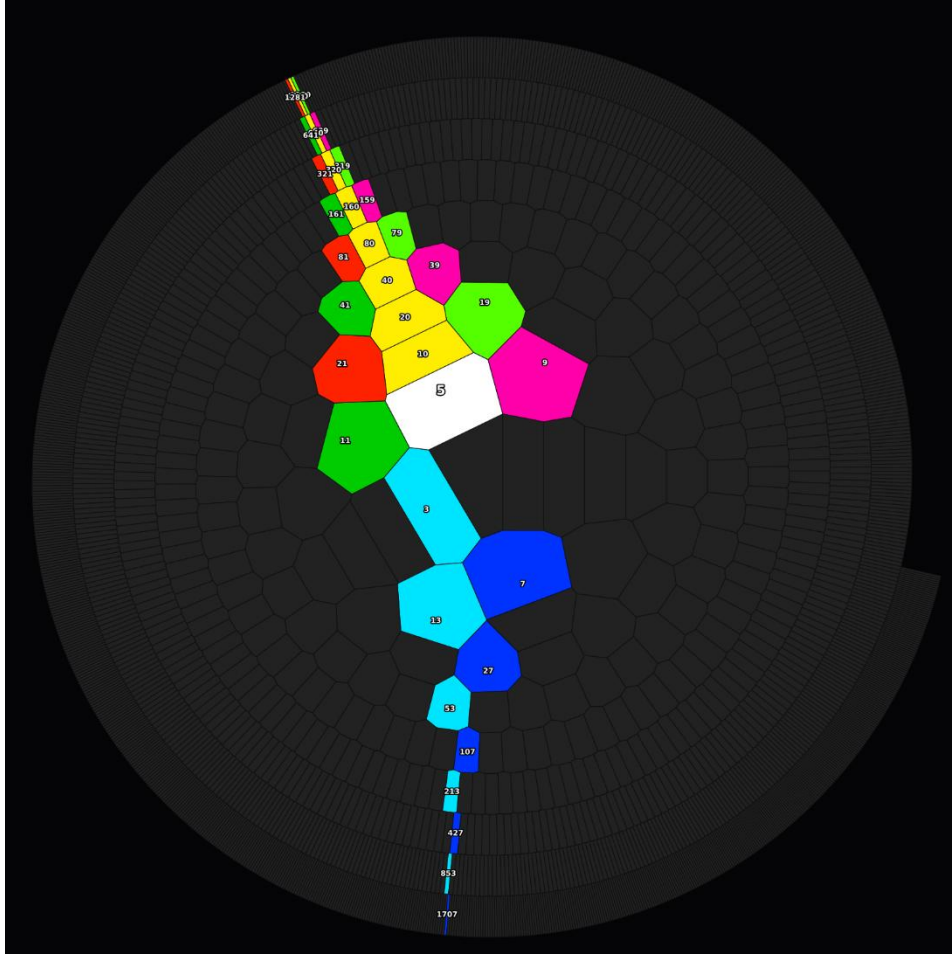


Figure 3. The $13 \rightarrow 5$ transition in Step 2 and the seven-system topology around the odd root $M = 5$.

It illustrates the geometric structure where the J_{even} sequence containing 13 $(3, 13, 53, \dots)$ transitions to the $5 \cdot 2^b$ axis—located between $3J$ and cJ —via the $3n + 1$ operation.

4.3 Step 3: $17 \rightarrow 13$ (Transition to the $13 \cdot 2^b$ Axis)

We generate the J sequence group containing 17.

$$17 = 9 + 8 = 9 + 4 \cdot 2$$

$$J(4, 9) = 4, 9, 17, 35, 69, \dots$$

$$3J = 12, 27, 51, 105, 207, \dots$$

$$cJ = 14, 25, 53, 103, 209, \dots$$

$$13 \cdot 2^b = 13, 26, 52, 104, 208, \dots = 13 \cdot 2^0, 13 \cdot 2^1, 13 \cdot 2^2, 13 \cdot 2^3, 13 \cdot 2^4, \dots$$

Therefore, the J_{even} terms of the 17-system (17, 69, ...) transition to the $13 \cdot 2^b$ axis via the $3n + 1$ operation (**Figure 4**). The integer 4 from J_{even} similarly transitions to the $13 \cdot 2^b$ axis, but being even, it undergoes division by 2 during the Collatz operation, eventually yielding 1.

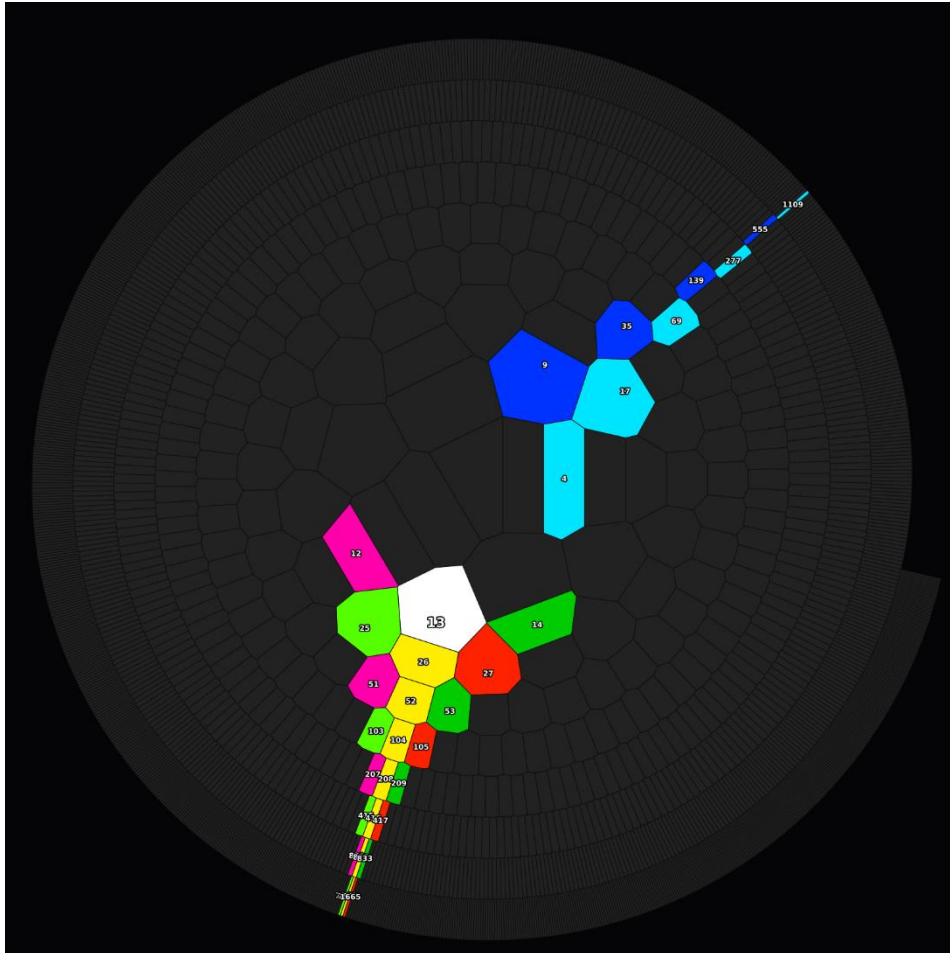


Figure 4. The $17 \rightarrow 13$ transition in Step 3 and the seven-system topology around the odd root $M = 13$.

It illustrates the geometric structure where the J_{even} sequence containing 17 (17, 69, ...) transitions to the $13 \cdot 2^b$ axis—located between $3J$ and cJ —via the $3n+1$ operation.

4.4 Step 4: 11 → 17 (Transition to the $17 \cdot 2^b$ Axis)

We generate the J sequence group containing 11.

$$11 = 6 + 5 = 6 + \frac{5}{2} \cdot 2$$

$$J\left(\frac{5}{2}, 6\right) = \frac{5}{2}, 6, 11, 23, 45, \dots$$

$$3J = \frac{15}{2}, 18, 33, 69, 135, \dots$$

$$cJ = \frac{19}{2}, 16, 35, 67, 137, \dots$$

$$17 \cdot 2^b = \frac{17}{2}, 17, 34, 68, 136, \dots = 17 \cdot 2^{-1}, 17 \cdot 2^0, 17 \cdot 2^1, 17 \cdot 2^2, 17 \cdot 2^3, \dots$$

Therefore, the integer terms of J_{even} from the 11-system (11, 45, ...) transition to the $17 \cdot 2^b$ axis via the $3n + 1$ operation (**Figure 5**).

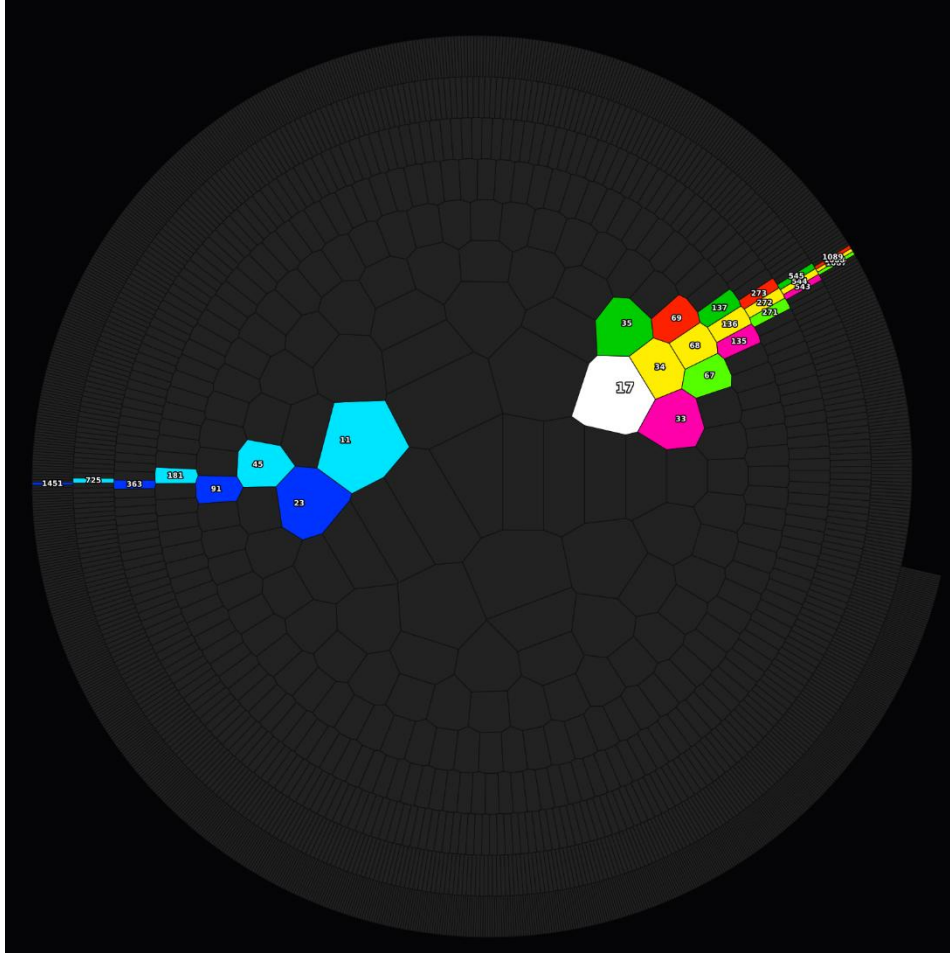


Figure 5. The 11→17 transition in Step 4 and the seven-system topology around the odd root $M = 17$.

It illustrates the geometric structure where the integer terms of the J_{even} sequence containing 11 (11, 45, ...) transitions to the $17 \cdot 2^b$ axis—located between $3J$ and cJ —via the $3n+1$ operation.

4.5 Step 5: $7 \rightarrow 11$ (Transition to the $11 \cdot 2^b$ Axis)

We generate the J sequence group containing 7.

$$7 = 4 + 3 = 4 + \frac{3}{2} \cdot 2$$

$$J\left(\frac{3}{2}, 4\right) = \frac{3}{2}, 4, 7, 15, 29, \dots$$

$$3J = \frac{9}{2}, 12, 21, 45, 87, \dots$$

$$cJ = \frac{13}{2}, 10, 23, 43, 89, \dots$$

$$11 \cdot 2^b = \frac{11}{2}, 11, 22, 44, 88, \dots = 11 \cdot 2^{-1}, 11 \cdot 2^0, 11 \cdot 2^1, 11 \cdot 2^2, 11 \cdot 2^3, \dots$$

Therefore, the integer terms of J_{even} from the 7-system (7, 29, ...) transition to the $11 \cdot 2^b$ axis via the $3n + 1$ operation (**Figure 6**). Through the example of the initial value 7, we have successfully described the odd Collatz orbit as a transition across J sequences.

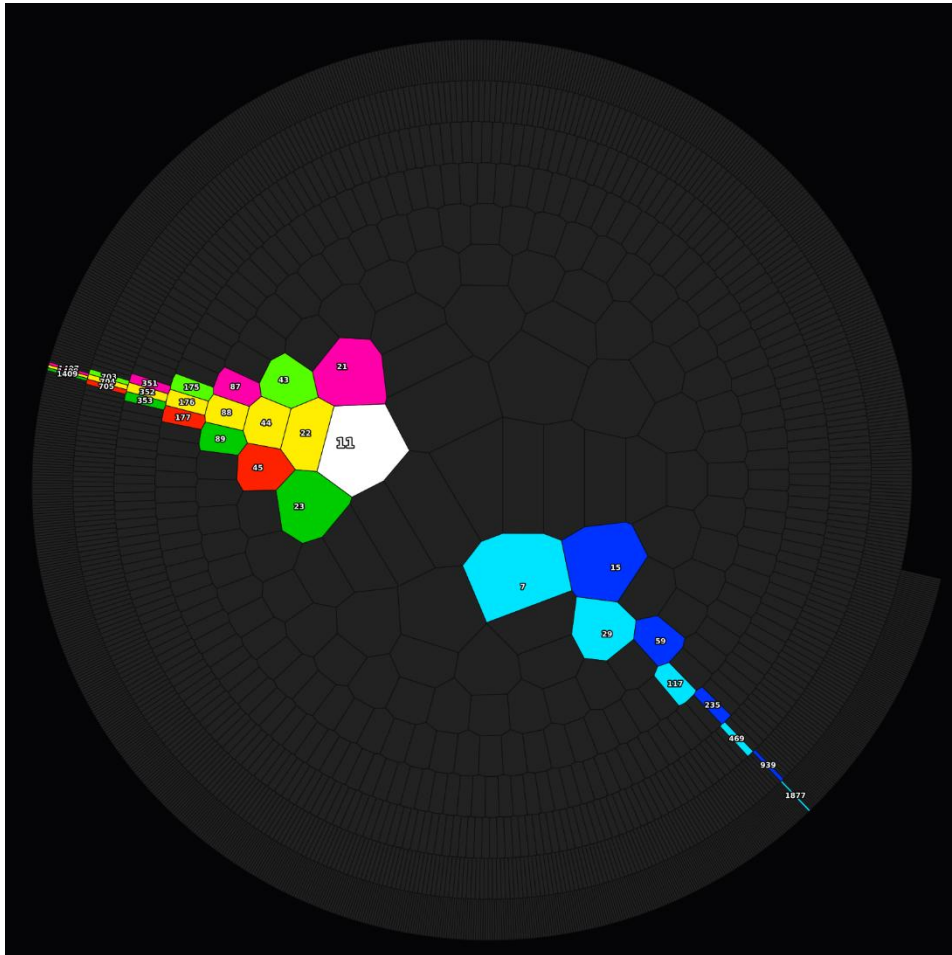


Figure 6. The $7 \rightarrow 11$ transition in Step 5 and the seven-system topology around the odd root $M = 11$.

It illustrates the geometric structure where the integer terms of the J_{even} sequence containing 7 (7, 29, ...) transitions to the $11 \cdot 2^b$ axis—located between $3J$ and cJ —via the $3n+1$ operation.

5. Results and Discussion

The visualization results confirm the following features:

- The seven-system sequences appear self-similarly around any arbitrary odd number M .
- The intersection points of the sequences geometrically align within the Voronoi space.
- The natural number space possesses a "stained-glass-like" geometric structure.

5.1 Self-similarity of the seven systems around arbitrary odd M

This is consistent with the fact that the local structure of the inverse Collatz tree contains J -type linear sequences.

5.2 Geometric alignment of intersection points in Voronoi space

In particular, the following intersection points form regular arrangements on the $\log_2$ spiral plane:

- $J \leftrightarrow 3J$
- $J \leftrightarrow cJ$
- $cJ \leftrightarrow 2^b$

5.3 Stained-glass geometry of the natural number space

Through Voronoi tessellation, it was visually confirmed that the regions occupied by natural numbers are densely packed without gaps. This reflects the fact that Voronoi partition inherently captures the dense packing structure of natural numbers in this topology.

6. Conclusion

This study provides a perspective that grasps the Collatz map not as a "sequence of arithmetic operations," but as a projection of the geometric and topological structures inherent in natural numbers. Specifically, the repeated appearance of the seven J -type systems around any arbitrary odd number M suggests that the natural number space possesses a highly ordered, resonance-based structure. Note that this study is based on observational visualization and does not constitute a rigorous mathematical proof.

Moving forward, this research can serve as a foundation for:

- Extension into p -adic spaces
- Geometric classification of inverse Collatz trees
- Information geometry studies of natural numbers

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